

выполним эквивалентное преобразование схемы и определим расчетные параметры цепи

$$\omega = 2\pi f = 2\pi \cdot 500 = 3141.5927 \text{ rad/s}$$

$$E_1 = \frac{E_{m1}}{\sqrt{2}} e^{j\varphi_1} = 29.698 e^{-j40^\circ} \text{ V}$$

$$J_2 = \frac{J_{m2}}{\sqrt{2}} e^{j\varphi_2} = 14.142 e^{j30^\circ} \text{ A}$$

$$E_2 = \frac{J_2}{G_{m2}} = \frac{12.25 + j7.07}{0.15} = 81.65 + j47.14 \text{ V}, R_{m2} = \frac{1}{G_{m2}} = \frac{1}{0.15} = 6.667 \Omega$$

$$E_{eq} = E_1 - E_2 = 22.75 - j19.09 - (81.65 + j47.14) = -58.9 - j66.23 \text{ V}$$

$$Z_{cг,м} = (R_{m1} + R_{m2}) + jX_{Lm1} = (3 + 6.667) + j6 = 14.67 + j6 \Omega = 15.85 \angle 22.25^\circ \text{ deg}$$

Рассчитаем эквивалентные сопротивления ветвей приемника

$$X_{L1} = \omega L_1 = 3141.5927 \cdot 0.0013 = 4.084 \Omega, X_{C1} = \frac{1}{\omega C_1} = \frac{1}{3141.5927 \cdot 10^{-6}} = 0 \Omega$$

$$X_{L2} = \omega L_2 = 3141.5927 \cdot 0.007 = 21.991 \Omega, X_{C2} = \frac{1}{\omega C_2} = \frac{1}{3141.5927 \cdot 0.000025} = 12.732 \Omega$$

$$X_{L3} = \omega L_3 = 3141.5927 \cdot 0.011 = 34.558 \Omega, X_{C3} = \frac{1}{\omega C_3} = \frac{1}{3141.5927 \cdot 0.0000056} = 56.841 \Omega$$

$$Z_1 = R_1 + jX_{L1} - jX_{C1} = 4 + j4.084 - j0 = 4 + j4.08$$

$$Z_2 = R_2 + jX_{L2} - jX_{C2} = 15 + j21.991 - j12.732 = 15 + j9.26$$

$$Z_3 = R_3 + jX_{L3} - jX_{C3} = 14 + j34.558 - j56.841 = 14 - j22.28$$

Находим эквивалентное сопротивление нагрузки и полное сопротивление цепи

$$Z_{23} = \frac{Z_2 Z_3}{Z_2 + Z_3} = \frac{(15 + j9.26)(14 - j22.28)}{29 - j13.08} = 14.58 - j0.51$$

$$Z_{cг,load} = Z_1 + Z_{23} = 4 + j4.08 + (14.58 - j0.51) = 18.58 + j3.58$$

$$Z_{in} = Z_{cг,м} + Z_{cг,load} = 14.67 + j6 + (18.58 + j3.58) = 33.25 + j9.58$$

Определим значения токов

$$I_1 = \frac{E_{eq}}{Z_{in}} = \frac{-58.9 - j66.23}{33.25 + j9.58} = -2.17 - j1.37 \text{ A}$$

$$I_2 = I_1 \frac{Z_3}{Z_2 + Z_3} = -2.17 - j1.37 \cdot \frac{14 - j22.28}{29 - j13.08} = -2.12 + j0.05 \text{ A}$$

$$I_3 = I_1 - I_2 = -2.17 - j1.37 - (-2.12 + j0.05) = -0.05 - j1.42 \text{ A}$$

Выполняем проверку расчетов по закону Кирхгофа

$$\Delta I = 0 + j0, \Delta U_1 = 0 + j0, \Delta U_2 = 0 + j0$$

$$\varepsilon_{KCL} = 0\%, \varepsilon_{KVL} = 0\%$$